Ando-type product norm inequalities
MathDB 342339

Principia Math in Progress

12 September 2026

Task 1. Resolve both proposed norm inequalities

Conjecture 2.1 of the published paper [1] (Conjecture 2.4 in its arXiv version) proposes two inequalities for every unitarily invariant norm. We prove both in the nonnegative function class used by its surrounding Theorems 2.1 and 2.2. The conjecture sentence omits nonnegativity in its operator-convex clause; we also resolve that literal signed version by an explicit counterexample below.

Theorem 1. *Let A, B be positive semidefinite complex $n \times n$ matrices and let $f(0) = 0$. If f is operator monotone on $[0, \infty)$, then*

$$|||(A - B)(f(A) - f(B))||| \le |||\,|A - B| f(|A - B|)\,||| . \tag{1}$$

If f is nonnegative and operator convex on $[0, \infty)$, then

$$|||(A - B)(f(A) - f(B))||| \ge |||\,|A - B| f(|A - B|)\,||| . \tag{2}$$

Both assertions hold for every unitarily invariant norm.

Write $s_1(X) \ge \cdots \ge s_n(X)$ for singular values and $\|X\|_{(k)} = \sum_{i=1}^{k} s_i(X)$ for the Ky Fan norms. By the Ky Fan dominance theorem, it suffices to prove each assertion for all k. The product $(A - B)(f(A) - f(B))$ need not be Hermitian.

The operator-monotone assertion

Put $C = A - B$, $H = f(A) - f(B)$, $x_i = s_i(C)$, and $y_i = s_i(H)$. Ando's classical norm inequality [3] gives

$$\sum_{i=1}^{j} y_i \le \sum_{i=1}^{j} f(x_i) \qquad (1 \le j \le n).$$

Here f is nonnegative because it is increasing and $f(0) = 0$. Set $z_i = f(x_i)$. The singular-value product inequality and Abel summation imply

$$\|CH\|_{(k)} \le \sum_{i=1}^{k} x_i y_i,$$

$$\sum_{i=1}^{k} x_i (y_i - z_i) = x_k \sum_{i=1}^{k} (y_i - z_i) + \sum_{j=1}^{k-1} (x_j - x_{j+1}) \sum_{i=1}^{j} (y_i - z_i) \le 0.$$

Since $x_i f(x_i)$ is decreasing in i, this proves $\|CH\|_{(k)} \le \||C|f(|C|)\|_{(k)}$, and hence (1). The product inequality follows by the von Neumann trace inequality applied to $\mathrm{Tr}(U^*CHV) = \mathrm{Tr}(HVU^*C)$, where U, V are $n \times k$ isometries: VU^*C has rank at most k and $s_i(VU^*C) \le s_i(C)$. Maximizing over U, V gives the claim.

If operator monotonicity is defined to permit a jump at the endpoint zero, the same argument follows by approximation. In that convention $f(t) = h(t) + a\mathbf{1}_{t>0}$, where $a \ge 0$ and h is continuous operator monotone with $h(0) = 0$. Apply the continuous case to $h(t) + at/(t+\varepsilon)$ and let $\varepsilon \downarrow 0$ in finite-dimensional functional calculus.

A weighted resolvent inequality

The following elementary argument extends the power weights in the trace method of [2] to spectral cutoffs.

Lemma 1. *Suppose D, E are commuting positive semidefinite matrices, with simultaneous diagonal entries d_i, e_i ordered in the same direction. If $W = W^*$ and $\|W\| \le 1$, then*

$$\|DWE\|_F^2 \le \|WDE\|_F^2 \le \mathrm{Tr}(D^2E^2),$$

where $\|\cdot\|_F$ is the Frobenius norm.

Proof. In a common eigenbasis, $|W_{ij}|^2 = |W_{ji}|^2$ and

$$\|WDE\|_F^2 - \|DWE\|_F^2 = \frac{1}{2}\sum_{i,j}(d_i^2 - d_j^2)(e_i^2 - e_j^2)|W_{ij}|^2 \ge 0.$$

The other inequality follows from $W^2 \le I$ by taking a trace against $D^2E^2 \ge 0$. □

Fix $\rho \ge 0$ and $0 \le \theta \le 1$, and define

$$q(t) = t\big(\mathbf{1}_{t>\rho} + \theta\mathbf{1}_{t=\rho}\big) \quad (t \ge 0). \tag{3}$$

Both $q(t)$ and $q(t)/(t+s)$ are nondecreasing for every $s > 0$.

Lemma 2. *If $B, C \ge 0$, $Q = q(C)$, and f is nonnegative operator convex with $f(0) = 0$, then*

$$\mathrm{Tr}\big(Q(f(B+C) - f(B))\big) \ge \mathrm{Tr}(Qf(C)). \tag{4}$$

Proof. For $s > 0$ put

$$g_s(t) = \frac{st}{s+t}, \qquad h_s(t) = \frac{st^2}{s+t} = st - sg_s(t).$$

Let $X = (C+sI)^{-1/2}$ and introduce the positive contractions

$$W = s(B+sI)^{-1}, \qquad V = (C+sI)^{1/2}(B+C+sI)^{-1}(C+sI)^{1/2}.$$

The resolvent identity gives

$$g_s(B+C) - g_s(B) = sWCXVX.$$

Using cyclicity of trace, set

$$M = XQ^{1/2}WC^{1/2}, \qquad N = C^{1/2}XVQ^{1/2}.$$

Then

$$\begin{aligned}\operatorname{Tr}\big(Q(g_s(B+C)-g_s(B))\big) &= s\operatorname{Tr}(MN)\\ &\le s\,\|M\|_F\,\|N\|_F \le s\operatorname{Tr}\big(QC(C+sI)^{-1}\big) = \operatorname{Tr}(Qg_s(C)).\end{aligned}$$

To justify the last bound, apply Lemma 1 first with $D = XQ^{1/2}$, $E = C^{1/2}$, and contraction W, and then with $D = C^{1/2}X$, $E = Q^{1/2}$, and contraction V. The two required monotone pairs of squared entries are respectively $q(t)/(t+s), t$ and $t/(t+s), q(t)$. Thus (4) holds for $h_s = st - sg_s$. It is equality for $f(t) = t$, and for $f(t) = t^2$ its difference is

$$\operatorname{Tr}\big(Q((B+C)^2 - B^2 - C^2)\big) = 2\operatorname{Tr}(QCB) \ge 0.$$

The standard integral representation of nonnegative operator-convex functions vanishing at zero is

$$f(t) = \beta t + \gamma t^2 + \int_{(0,\infty)} h_s(t)\,d\mu(s), \qquad \beta,\gamma \ge 0, \quad \mu \ge 0. \tag{5}$$

One obtains it by applying the positive operator-monotone representation to $f(t)/t$; its value at zero is the nonnegative right derivative β. This is the representation used in [1]. Integrating the established inequalities proves (4). Truncated integrals pass to the full integral by finite-dimensional functional calculus, since the scalar representation converges at every eigenvalue. □

Lemma 3. *If $T, D, Q \ge 0$, $QD = 0$, and f is as in Lemma 2, then*

$$\operatorname{Tr}\big(Q(f(T) - f(T+D))\big) \ge 0. \tag{6}$$

Proof. For the linear and quadratic functions the trace in (6) vanishes: every term contains QD or DQ after a cyclic permutation. For h_s it equals

$$s\operatorname{Tr}\big(Q(g_s(T+D) - g_s(T))\big) \ge 0.$$

Indeed, g_s is operator monotone, as follows directly from inverse order and $g_s(T) = sI - s^2(T+sI)^{-1}$. Now integrate using (5). □

The operator-convex assertion

Let $C = A - B = C_+ - C_-$, with $C_+C_- = 0$, and put $Z = A + C_- = B + C_+$. For each k, choose a cutoff ρ and a fraction θ so that

$$P = \mathbf{1}_{|C|>\rho} + \theta\mathbf{1}_{|C|=\rho}, \qquad 0 \le P \le I, \qquad \operatorname{Tr} P = k.$$

For example, take ρ to be the kth largest eigenvalue of $|C|$ and divide the remaining required trace equally across its eigenspace. This also covers zero eigenvalues and ties. With q from (3), put $Q_+ = q(C_+)$ and $Q_- = q(C_-)$. Then $PC = Q_+ - Q_-$ and $Q_+C_- = Q_-C_+ = 0$. Consequently,

$$\begin{aligned}\operatorname{Tr}\big(PC(f(A) - f(B))\big) &= \operatorname{Tr}\big(Q_+(f(A) - f(Z))\big) + \operatorname{Tr}\big(Q_+(f(Z) - f(B))\big)\\ &\quad + \operatorname{Tr}\big(Q_-(f(Z) - f(A))\big) + \operatorname{Tr}\big(Q_-(f(B) - f(Z))\big)\\ &\ge \operatorname{Tr}(Q_+f(C_+)) + \operatorname{Tr}(Q_-f(C_-))\\ &= \operatorname{Tr}\big(P|C|f(|C|)\big).\end{aligned}$$

The first and fourth terms are nonnegative by Lemma 3; the second and third are bounded by Lemma 2. Since f is nonnegative convex and $f(0) = 0$, $tf(t)$ is nondecreasing. Thus the last trace equals $\||C|f(|C|)\|_{(k)}$.

For every matrix Y, the von Neumann trace inequality gives $|\operatorname{Tr}(PY)| \le \|Y\|_{(k)}$ whenever $0 \le P \le I$ and $\operatorname{Tr} P = k$: the singular values of P lie in $[0, 1]$ and sum to k. Applying this to $Y = C(f(A) - f(B))$ proves

$$\|C(f(A) - f(B))\|_{(k)} \ge \||C|f(|C|)\|_{(k)}.$$

Ky Fan dominance proves (2) and completes the theorem.

The omitted sign assumption

Without nonnegativity, the operator-convex clause is false already for scalars. Take $f(t) = t^2 - 3t$, $A = 2$, and $B = 1$. This f is operator convex and $f(0) = 0$, but $f(2) = f(1) = -2$, so its proposed inequality reads $0 \ge 2$ in the absolute-value norm. The theorem above supplies the substantive nonnegative version, as well as settling the literal statement's scope.

Task 2. Classify equality in the norm inequalities

Determine equality cases for individual Ky Fan and Schatten norms, tracking equality in the resolvent contractions and the spectral cutoff trace bound.

Task 3. Classify admissible weights

Determine which scalar weights q preserve the ordered inequality (4), and whether monotonicity of q and $q(t)/(t + s)$ for all $s > 0$ is necessary for a universal statement.

Task 4. Find the maximal function classes

Characterize functions satisfying (1) or (2) for all positive matrices and all unitarily invariant norms, and decide how far the operator-monotone and operator-convex hypotheses can be weakened.

Task 5. Extend the inequalities to operator ideals

Establish analogues for compact operators and semifinite noncommutative integration, specifying the integrability and approximation assumptions needed for singular-value dominance.

References

[1] T. H. Dinh, M. T. Ho, C. T. Le, and B. K. Vo, *Two trace inequalities for operator functions*, Mathematical Inequalities & Applications 22 (2019), 1021–1026. Published paper, Conjecture 2.1; arXiv:1904.01961v1, Conjecture 2.4.

[2] T. H. Dinh, R. Dumitru, J. A. Franco, and C. T. Le, *On some trace inequalities*, Mathematical Inequalities & Applications 23 (2020), 467–476. Published paper.

[3] T. Ando, *Comparison of norms* $|||f(A) - f(B)|||$ *and* $|||f(|A - B|)|||$, Mathematische Zeitschrift 197 (1988), 403–409.